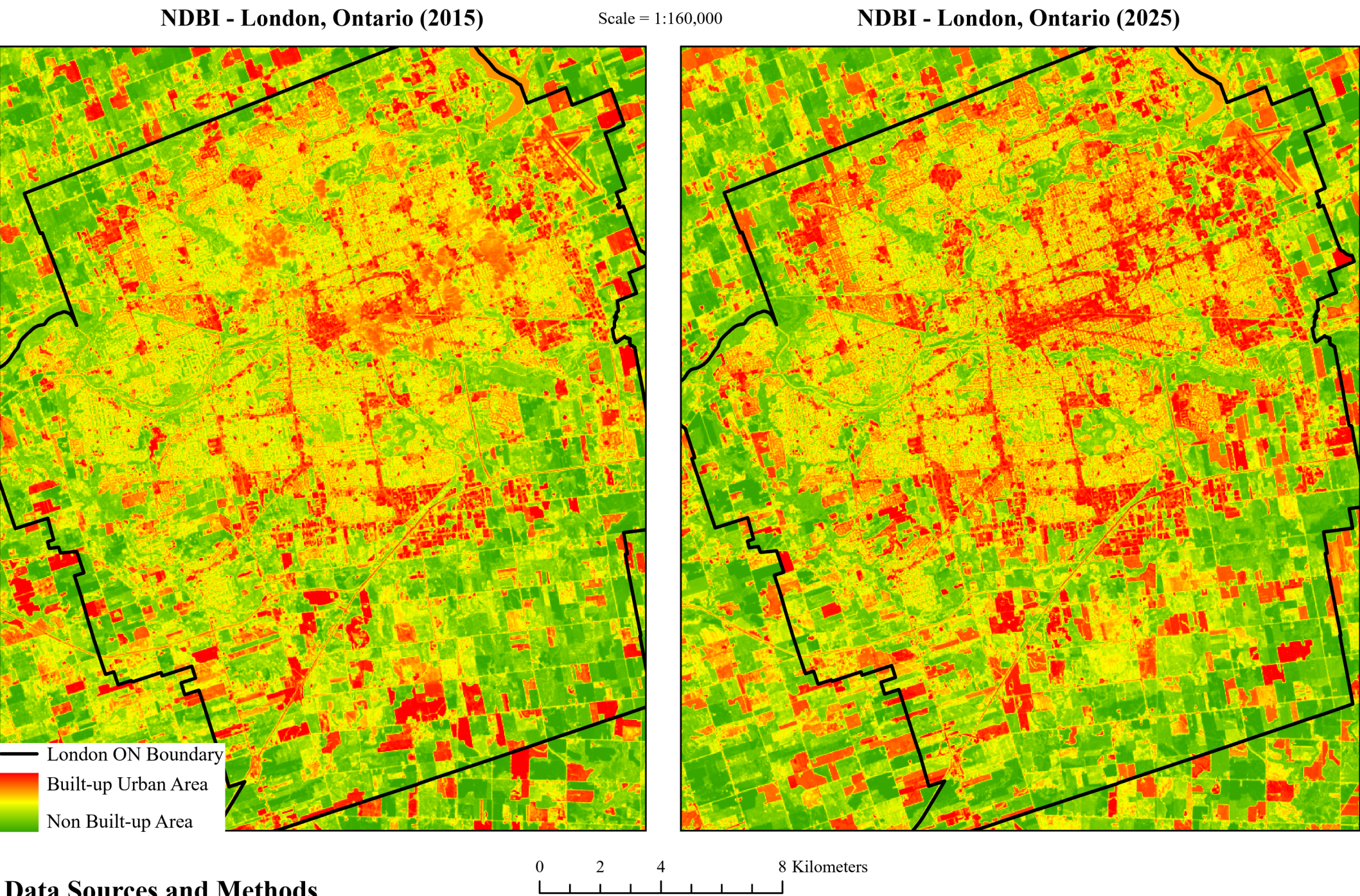


Urban Growth Analysis in London, Ontario (2015–2025) Using NDBI



Data Sources and Methods

Satellite imagery was obtained from the United States Geological Survey (USGS) EarthExplorer platform using Landsat 8 (2015) and Landsat 9 (2025) Operational Land Imager (OLI) data. Images acquired on July 28, 2015 and July 15, 2025 were selected to ensure consistent seasonal conditions and minimize variation in vegetation and illumination. Both datasets correspond to WRS-2 Path 19 and Row 30, covering the City of London, Ontario.

Normalized Difference Built-up Index (NDBI) was calculated for each year using Bands 6 (Shortwave Infrared, SWIR) and 5 (Near Infrared, NIR), following the equation: $NDBI = (SWIR - NIR) / (SWIR + NIR)$. The Shortwave Infrared (SWIR) band is sensitive to moisture content and is strongly reflected by built-up and impervious surfaces, while the Near Infrared (NIR) band is strongly reflected by healthy vegetation. As a result, built-up areas produce higher (positive) NDBI values, while non built-up and natural surfaces produce lower (negative) values, allowing for effective differentiation between urban and non-urban land cover.

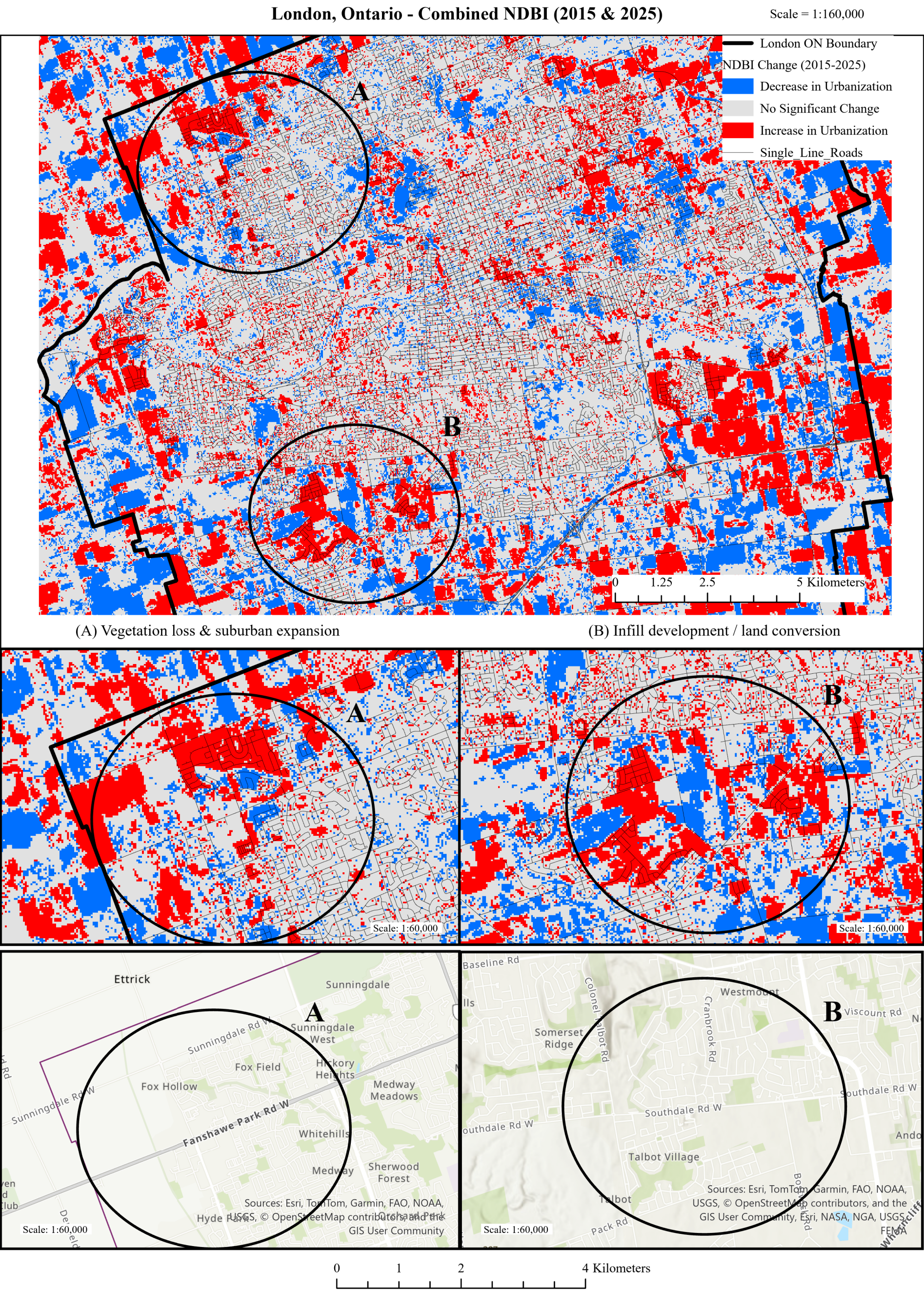
Raster calculations were performed in ArcGIS Pro, and resulting NDBI layers were clipped to the City of London boundary using municipal open data. All datasets were projected to WGS 1984 / UTM Zone 17N. A change detection raster was created by subtracting the 2015 NDBI from the 2025 NDBI to identify areas of urban growth and decline. To improve interpretability, the change raster was reclassified into three categories: increase in urbanization, no significant change, and decrease in urbanization. These classes were symbolized using a red–grey–blue colour scheme. To provide spatial context, a basemap and city boundary were included. Areas of significant change were identified and highlighted using labelled polygons (A and B), and inset maps were created to provide detailed views of these locations. Additional contextual imagery (including map and street-level views) was incorporated to support interpretation of land use change at specific sites.

Results

The NDBI results indicate a clear increase in built-up land within London, Ontario between 2015 and 2025. The 2025 map shows higher NDBI values along the urban periphery, reflecting expansion of impervious surfaces such as residential developments and infrastructure. In contrast, the 2015 imagery presents more extensive vegetated and open land surrounding the city, which is consistent with lower NDBI values.

The classified change map further highlights these patterns, with red areas indicating significant increases in built-up surfaces and blue areas representing localized decreases or variations in land cover. The central urban area shows relatively little change, as it was already highly developed in 2015, while the most notable changes occur along the city’s outer edges.

The highlighted areas (A and B) clearly illustrate these trends. In Area A, located along the northern urban boundary, the imagery shows a transition from predominantly natural or agricultural land in 2015 to developed suburban land in 2025, indicating outward expansion. Similarly, Area B demonstrates substantial growth, with large patches of previously non-built-up land converted into urban surfaces. The inset maps provide a more detailed view of these transformations, confirming patterns of suburban expansion and land conversion. Overall, the results demonstrate a pattern of outward urban growth, with development concentrated along the city’s periphery. The combination of NDBI analysis, classification, and focused inset areas provides a clear and interpretable representation of urban expansion and land use change over the ten-year period in London between 2015 and 2025. This change is primarily characterized by outward suburban expansion rather than internal densification.



Normalized Difference Built-up Index (NDBI)

$$NDBI = (SWIR + NIR) / (SWIR - NIR)$$